

SKILL WEEK 14

Task 1:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int p1[2], p2[2];
    pipe(p1);
    pipe(p2);

    if (fork() == 0) {
        dup2(p1[1], STDOUT_FILENO);
        close(p1[0]); close(p1[1]); close(p2[0]); close(p2[1]);
        execlp("ls", "ls", NULL);
        return 1;
    }

    if (fork() == 0) {
        dup2(p1[0], STDIN_FILENO);
        dup2(p2[1], STDOUT_FILENO);
        close(p1[0]); close(p1[1]); close(p2[0]); close(p2[1]);
        execlp("grep", "grep", ".c", NULL);
        return 1;
    }

    if (fork() == 0) {
        dup2(p2[0], STDIN_FILENO);
        close(p1[0]); close(p1[1]); close(p2[0]); close(p2[1]);
        execlp("wc", "wc", "-l", NULL);
        return 1;
    }

    close(p1[0]); close(p1[1]); close(p2[0]); close(p2[1]);
    wait(NULL); wait(NULL); wait(NULL);

    printf("Large pipeline completed successfully.\n");
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 14.1.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 14.1.c -o 14.1
ashishreddy@Ashishs-MacBook-Air ~ % ./14.1
18
Large pipeline completed successfully.
```

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Task 2:

Code:

Project Architecture

```
User
|
v
Command Parser
|
+----> Built-ins
|
+----> Pipeline Manager
|       |
|       +----> fork()
|       +----> pipe()
|       +----> dup2()
|
+----> Job Manager
|       |
|       +----> Signals
|       +----> Process Groups
```

Execution uses Linux/POSIX system calls.

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano architecture.txt
ashishreddy@Ashishs-MacBook-Air ~ % cat architecture.txt
Project Architecture
```

```
User
|
v
Command Parser
|
+----> Built-ins
|
+----> Pipeline Manager
|       |
|       +----> fork()
|       +----> pipe()
|       +----> dup2()
|
+----> Job Manager
|       |
|       +----> Signals
|       +----> Process Groups
```

Execution uses Linux/POSIX system calls.

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Task 3:

Code:

```
CC=gcc
CFLAGS=-Wall -Wextra

all: shell

shell: main.c
■$(CC) $(CFLAGS) main.c -o shell

clean:
■rm -f shell

run: shell
■./shell

README:
■echo "# OSSP Project" > README.md
■echo "Build: make" >> README.md
■echo "Run: make run" >> README.md
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano Makefile
ashishreddy@Ashishs-MacBook-Air ~ % make
gcc -Wall -Wextra main.c -o shell
ashishreddy@Ashishs-MacBook-Air ~ % make README
ashishreddy@Ashishs-MacBook-Air ~ % cat README.md
# OSSP Project
Build: make
Run: make run
```

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Task 4:

Code:

```
#!/bin/bash

echo "=== OSSP FINAL DEMO ==="
echo "1. Built-in command test"
echo "2. Pipeline test"
echo "3. Input/output redirection test"
echo "4. Background job test"
echo "5. SIGINT and SIGTSTP test"
echo "6. Job table test"
echo "All demonstration scenarios prepared."
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano demo.sh
ashishreddy@Ashishs-MacBook-Air ~ % chmod +x demo.sh
ashishreddy@Ashishs-MacBook-Air ~ % ./demo.sh
=== OSSP FINAL DEMO ===
1. Built-in command test
2. Pipeline test
3. Input/output redirection test
4. Background job test
5. SIGINT and SIGTSTP test
6. Job table test
All demonstration scenarios prepared.
```